

Blockchains, Smart Contracts and Connected Sensors: Enforceable Execution vs Better Information *

Yannis Bakos[†]

Hanna Halaburda[‡]

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Abstract

Smart contracts as automation existed before blockchain technology, but blockchains have been a key reason for their recent popularity and their emerging use in areas such as Decentralized Finance (DeFi). The key feature of blockchain-based smart contracts is that they guarantee execution if their conditions are met. Unlike traditional off-chain smart contracts where the parties may renege and compliance may require institutional enforcement, such as recourse to litigation and the courts, compliance in blockchain-based smart contracts is assured by the applicable protocols. In addition, smart contracts critically depend on digital inputs that inform them whether their triggering conditions have occurred, typically provided by connected sensors (such as “IoT”) that supply certified signals to the blockchain. If there is a dispute, evidence from these sensors can be shown to a court or an arbitrator, making the corresponding states “verifiable.” The two technologies are thus intricately linked but have distinct functionality; in fact much of the purported value of employing smart contracts in the popular press upon closer examination comes from the associated use of connected sensors.

We develop a model for blockchain-based smart contracts that distinguishes their impact from that of connected sensors. Smart contracts restrict the strategy space by allowing the contracting parties to commit not to hold-up each other; this typically increases the contracting region where trade occurs. Connected sensors expand the state space over which a contract can be specified by creating finer partitions of the verifiable states of nature. We show that when the cost of contract enforcement is low compared to the potential gains from trade, the ability of blockchain-based smart contracts to prevent renegeing adds little value, as the contractual terms can be enforced at a low cost with alternative mechanisms. When the cost of contract enforcement is comparable to the potential gains from trade, there will be no trading without either smart contracts to reduce the enforcement cost, or connected sensors to increase the value generated at equilibrium. Finally, when the cost of contract enforcement is high compared to the potential gains from trade and the costs of technology implementation, trade will not take place without smart contracts.

Keywords: smart contracts, connected sensors, Internet of Things, IoT, blockchain

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[†]Stern School of Business, New York University, bakos@stern.nyu.edu

[‡]Stern School of Business, New York University, hhalaburda@gmail.com

1 Introduction

It is often argued that blockchain technology can transform interactions between firms by enabling governance based on “smart” contracts rather than the legal system (e.g., see Werbach and Cornell (2017), Werbach (2018), Halaburda et al (2019), Lumineau et al (2021), Halaburda et al (2022)). The key characteristic of smart contracts, a concept introduced by Nick Szabo (1996), is their automated algorithmic execution based on a mapping from certain detectable states of the world to corresponding actions. The increased attention to applications of blockchain-related technologies has brought in focus the potential economic role of smart contracts, with frequent claims that they will improve economic efficiency, while also potentially affecting firm boundaries and business models.

While smart contracts have existed for a long time without blockchains (e.g., limit brokerage orders and vending machines), the increasing prevalence of blockchain technologies can broaden the scope and applicability of smart contracts. Specifically, blockchains provide an infrastructure for the recording and execution of smart contracts, certify the occurrence of contracted states of the world through technologies such as connected sensors and blockchain oracles,¹ provide execution of certain actions such as a payment transaction “on the blockchain” thus assuring their irreversibility (e.g., see Holden and Malani 2021; Gans 2021), and can relay required actions to off-chain platforms for execution. For instance the emergence of blockchains as platforms for smart contracts has been a key enabler of using smart contracts to govern short-term interactions among potentially anonymous parties that do not know and do not trust each other and thus cannot rely on the legal system for enforcement, such as Decentralized Finance (DeFi).

In discussing the role of smart contracts, the literature draws on examples dating back to Szabo’s proposal for a “smart lien” on a car, which would automatically “return control of the car keys to the bank” if the borrower fails to make payments and thus “might be much cheaper and more effective than a repo man.” More recent examples include using real-time transaction data to provide automated sales financing (Tinn 2018, 2020) and taking automatic actions when a transported good is subjected to certain events, which is one of the capabilities of IBM Food Trust, the blockchain-enabled food supply chain platform from IBM Supply Chain Intelligence Suite. For instance, Food Trust can trigger automatic shipment of replacement for fruit that has not been properly refrigerated while in transit; notably, this can take place as soon as the transported fruit is exposed to abnormal temperature, without waiting for it to be inspected upon delivery.

The above examples illustrate that smart contracts critically depend on digital inputs that in-

¹A blockchain oracle is a service that provides smart contracts with information from the outside world. It authenticates, queries and verifies external data sources, usually via trusted APIs, and then relays the relevant information to smart contracts.

form them that a certain state of the world has occurred, so that they can trigger the corresponding actions. These inputs may already exist in digital form, such as digital notification of a delivery, or a trade execution, or a missed payment, but some of the most promising applications of smart contracts rely on new sources of digital information. In the smart lien example, the smart contract not only needs to be informed of the failure to make payments, but also connected sensors would be needed that can disable the car on the bank's command and can communicate the car's location to the bank.

Predictions for the coming prevalence of smart contracts thus should be closely linked to the availability of connected sensors (like those that are part of IoT—the “Internet of Things”) and of blockchain oracles. Such sensors and oracles increase the ability to identify states of the world with high accuracy and at a fine-grained level of discrimination, and thus can render such states “verifiable” as they provide evidence that can be shown to a court or an arbitrator in the event of a dispute. This would allow contracting based on the occurrence of these newly verifiable states, which could support higher value outcomes, for instance by incentivizing efficient but otherwise costly actions.² For simplicity we use the term “connected sensors” to describe technologies that can provide verifiable and actionable digital inputs to smart contracts, which includes blockchain oracles even if they are not mentioned explicitly.

Connected sensors enable increased detail in both “conventional” and “smart” contracts, and can add significant value on their own, yet popular accounts for the importance of smart contracts often confound the implications of the two technologies. For instance, in Szabo's foundational example of the smart lien, arguably most of the value is created by the ability to determine the location of the car and remotely disable it, which is provided by connected sensors, rather than by the automatic execution provided by the smart contract. On the other hand, for a payment-upon-delivery system in an international trade context where the cost of conventional arrangements based on legal rules, posting bonds or arbitration would be high, most of the value may be created by the irreversible automatic execution provided by the smart contract, because it eliminates the risk of non-payment by the recipient.

The introduction of smart contracts and connected sensors changes the strategic interaction between the contracting parties. This is not just an incremental quantitative change in the value of certain parameters, such a reduction in a cost component, but a qualitative change in the structure of the interaction that is different for each of these technologies; we formalize these structural changes using game theory. To understand the impact of these changes, we develop a simple modular framework where we tractably introduce each of these technologies as well as their

²Certifying the information provided by sensors and oracles to make the identified states “verifiable” will likely present a tradeoff between cost and degree of verifiability, which will need to be addressed to the satisfaction of the contracting parties.

combination. We then derive a set of results that help us understand the interplay between the different technology options and the resulting outcomes. We explore these questions by modeling connected sensors and smart contracts in a simplified supply chain setting where a perishable good (“fruit”) is transported requiring a costly action by the transportation company (“refrigeration”).

We characterize the implications of automated and irreversible execution that can be provided by blockchain-based smart contracts, the implications of more granular verifiable states that can be identified by connected sensors, and how they each affect the scope and efficiency of contracting. In our analysis we consider a smart contract that triggers irreversible payment upon delivery; this could be achieved either via direct blockchain verification, such as a digital token representing proof of delivery, via an appropriately verified sensor reading, such as scanning a tamper proof tag on a shipping container, or via a blockchain oracle such as a system that obtains verified information from port authority or customs records and provides a corresponding indication of delivery on the blockchain.

We also consider the implications of connected sensors that can verifiably determine and communicate on the blockchain whether proper refrigeration was provided during transportation; this could be accomplished, for instance, by sensors that indicate whether transport temperature was kept in the appropriate range, are able to detect tampering with either the container or the sensors, and can communicate their readings to the blockchain either directly or via a trusted oracle. Implementing this type of sensor makes refrigeration during transport verifiable, as the readings of the sensors can be provided to a third party such as a court or an arbitrator, and also allows the contracted payment upon delivery to be contingent on whether proper refrigeration was provided.

Our analysis shows that the cost of contract enforcement (e.g., the cost of legal action or arbitration) is a key parameter in our setting. This cost can be substantial; for instance, in our setting involving international transportation, it would likely be prohibitively expensive to resolve disputes in court. Arbitration might be cheaper, but it would still be very expensive.³ Even if the losing party is obligated to reimburse the arbitration costs of the winning party, the uncertainty of resolution, delay of over a year in settling the dispute and uncertainty in being able to collect any award may represent a significant cost.⁴ Similarly, in settings with short term interactions among anonymous participants, such as blockchain-based decentralized business platforms like dApps or

³A 2015 report about the London Court of International Arbitration (one of four international arbitration courts) found that the median cost of arbitration is US\$99,000 and mean is US\$192,000; the median duration of LCIA arbitrations is between 15 and 19 months and the mean between 18.5 and 21 months; cf. LCIA (2015), Morgan Lewis (2015). Newing et al (2019) report that litigation in court is longer, more expensive, and on average the winning party recovers only 60% of the costs from the losing party.

⁴Mechanisms like letters of credit typically cost a few percent of the total transaction cost, but involve several months of delay before payment is processed, may not be available to entities without financial resources or banking history, and while they address credit concerns, in order to protect from reneging they typically require recourse to courts or arbitration.

DeFi, participants do not know and do not trust each other and cannot rely on the legal system or arbitration mechanisms for enforcement, and thus the cost of enforcing agreements is practically infinite.

We show that when the cost of contract enforcement is low compared to the potential gains from trade, the ability of smart contracts to prevent reneging adds little value, as the contractual terms can be enforced at a low cost with alternative mechanisms. Connected sensors making relevant states verifiable are beneficial, as long as their cost is lower than the resulting gain from enabling a higher value equilibrium outcome.

When the cost of contract enforcement is comparable to the potential gains from trade, there will be no trading without either smart contracts to reduce the enforcement cost, or connected sensors to increase the value generated at equilibrium. Depending on their implementation cost, both smart contracts and connected sensors can add value, but depending on the parameter values, it may be efficient to employ one but not both.

When the cost of contract enforcement is high compared to the potential gains from trade and the costs of technology implementation, trade will not take place without smart contracts, and sensors can improve the contracting outcome as long as their cost is lower than the gain from the higher value equilibrium. In this case connected sensors create value only if both technologies can be implemented together.

Our contribution in this work is to characterize when it would be individually and socially optimal to implement only smart contracts (based on limited existing digital data), only connected sensors (providing additional verifiable information), or both sensors and smart contracts (i.e., where synergies exist between them). Furthermore, we show that the incentives to adopt these two technologies may differ for different economic agents, and may not be aligned with social optimality. Our results also demonstrate that smart contracts may “democratize” certain business areas by providing alternatives to other mechanisms to operate in environments with high enforcement costs, such as relational contracting, reputation or bonding, all of which tend to favor established or large participants.

Our findings can guide investment decisions in smart contracts and connected sensors by investors and platforms, or policy and subsidization decisions by governments. They also illustrate the key role of smart contracts in enabling the emerging blockchain-based decentralized business models such as DeFi that would not be able to function without the commitment provided by smart contracts.

2 Related Literature

While blockchain-based smart contracts are a relatively recent phenomenon, they have sparked an important stream of research in the rapidly growing blockchain literature. Tinn (2018) investigates how the new functionality afforded by smart contracts could improve the efficiency of financing contracts. Cong and He (2019) study the impact of smart contracts on the competitive environment and find that the effect of the technology on welfare is ambiguous. Depending on the environment and use of smart contracts, they may facilitate firm entry and enhance competition and welfare; but they may also help incumbent firms to prevent new entrants, thus perpetuating oligopolies to the detriment of social welfare. Gans (2021) looks at more general implications of smart contracts. He shows that by improving observability and reducing the costs of verifying the performance of contractual obligations, the space of feasible contracts can be enlarged. He concludes that in order to achieve either, smart contracts need verifiable inputs originating outside the blockchain network on which they are implemented.

Chod et al (2020) develop a model showing that in a setting of financing with inventory as collateral, a combination of IoT sensors and smart contracts on a permissionless blockchain leads to more efficient market outcomes as a result of verifiable information on inventory levels, but only if the cost of obtaining such verifiable signals is low enough. They also develop a protocol—based on Bitcoin’s network—that would allow for such low-cost verifiability. Shibuya and Babich (2021) consider supply chain financing in a setting where the combination of sensors and blockchain technology enables verifiability of collateral assets held by higher-tier players, not just the ones that interact directly with the borrower. They find that this extended verifiability may have a detrimental effect, in which case the traditional financing process would be preferred, demonstrating that more information about the occurring states of the world does not necessarily lead to more efficient outcomes.

The potential impact of smart contracts has also been analyzed in the law literature. Werbach and Cornell (2017) investigate the popular claim that smart contracts could replace contract law and the need for courts to adjudicate contractual disputes. As they investigate both the potential and the limitations of smart contracts, they conclude that while smart contracts will require new legal responses, they will not displace contract law. Casey and Niblett (2018) envision more advanced forms of smart contracts than what is currently available, with smart contracts that utilize Artificial Intelligence to dynamically fill in the gaps that may have been overlooked in the initial agreement, without direct involvement of the contracting parties. This would substitute for courts filling these gaps ex-post and raise the question of what changes in the doctrine and theory of contract law would be needed to account for this possibility. Holden and Malani (2021) discuss how the

automatic execution of terms in smart contracts on blockchain networks can help to overcome the hold-up problem by either preventing renegotiation of agreements or providing a structure within which such renegotiation can take place. They argue that smart contracts may thus provide a tool to implement theoretical mechanisms that increase the efficiency of contracting, but which have been difficult to execute in practice. They postulate that for this to work, all relevant data and assets must be available on the same blockchain platform that supports the smart contract, and that the core function of blockchain networks is “witnessing,” i.e., the provision of authoritative and publicly observable information based on appropriate cryptographic algorithms.

The above papers address the importance of digital inputs corresponding to relevant states of the world for the implementation of smart contracts. In that context, they frequently mention the need to provide connected sensors and the important role of the Internet of Things in that. They do not explore, however, how much of the proposed benefit of smart contracts comes from the sensors alone, and do not consider substitution and complementarities between these technologies.

Cong and He (2019) specifically focus on how the distributed consensus of the blockchain affects the information about the state of the world, and thus the contract terms that will be executed. They consider a situation where multiple validators have access to IoT sensors, and therefore obtain readings on the state of the world. Similarly to us, Cong and He assume that IoT sensors always provide correct readings. They focus on the incentives of the validators to misreport the sensor readings, and the resulting need for a large number of players to have access to the information about the state of the world.

Our approach differs as we adopt the approach that reports from oracles are digitally signed and cannot be blocked or misreported (e.g., Zhang et al 2016); instead we focus on how smart contracts and connected sensors change the structure of the contractual space in our setting, and the resulting changes in economic outcomes.

Gans (2021) also recognizes the importance of verifiable digital inputs for applications of smart contracts and agrees that blockchain systems can provide verifiable information in certain cases, but he argues that typically what the blockchain can provide is not enough. Thus he finds limited appeal in smart contracts and instead searches for mechanisms to optimize trading given the impossibility of verifiable digital information. As an example, he sets up a clever multi-step mechanism that achieves efficient trade without verifiable digital information, based on the information voluntarily revealed by the parties.

As before, our approach differs in that we focus on the case where IoT sensors can provide reliable and verifiable digital inputs that can be observed and trusted by all parties, and where these inputs can be made available to smart contracts.⁵ We also recognize that introducing these

⁵We believe that in reality this is a question of degree; there is a tradeoff between the strictness of requirements

connected sensors offers benefits (and imposes costs) by itself, before smart contracts are added. In this paper we distinguish the impact of the information provided by the connected sensors from the impact of automated execution afforded by the smart contracts, and we formalize how each technology affects the contracting game, the strategy space, and the resulting equilibria.

Finally, given the question we study, established tools of principal-agent theory and contracting theory are directly relevant. Connected sensors affect the observability of the states of the world, and offer the possibility to account for more such states in the contract, which has implications that go back to Spence and Zeckhauser (1971), and Holmström (1979). The core of smart contracts is automated execution, which relates to credible commitment, e.g., as in Schelling (1960), and the hold-up problem, e.g., as in Grossman and Hart (1986) and Hart and Moore (1988).⁶

For the purposes of our analysis, a simpler setting than most of this literature suffices: we assume no uncertainty and thus we do not need to consider risk preferences and asymmetric information; however we do consider moral hazard. The contracting problem that still remains is the one of contract enforcement, or providing “proper incentives for honoring agreements” as put by Greif (1993). In addition to contract enforcement via legal action, there is a literature going back to Greif (1993) and Tirole (1996) that considers alternative mechanisms based on social enforcement such as relational contracts and reputation.

3 Model Setup

We model contracting to trade in a setting similar to Holden and Malani (2021) and Gans (2021). Specifically, we consider a principal F that desires to transport a perishable good to which we refer as “fruit.” Transportation is provided by an agent T that has a costly action, to which we refer as “refrigeration,” that affects the quality of the fruit upon delivery and thus the economic value generated. If the fruit is shipped under proper refrigeration, it is of high quality when delivered, will last longer on the shelf, and provide higher utility to the end consumers. If not properly refrigerated, the fruit will deteriorate faster in condition and taste, and may even spoil before it can be sold. In the base case, F will eventually obtain a (possibly noisy) signal of whether the fruit was refrigerated, but refrigeration is not verifiable in the sense that F cannot prove to a third party whether the fruit was refrigerated in transit.

More formally, the value of the delivered fruit to F is v_q , where $q \in \{L, H\}$ denotes low (without refrigeration) or high (with refrigeration) quality of transportation and T ’s cost is c_q ,

placed on the verifiability of “off-chain” inputs, and the feasibility and cost of obtaining them and connecting them to smart contracts.

⁶Maskin and Tirole (1999) argue that complex contracts can solve the hold-up problem when there are ex-ante indescribable contingencies, while Hart and Moore (1999) counter-argue that this solution does not work when renegotiation cannot be ruled out.

with $v_q > c_q > 0$. If the fruit is not transported, both parties obtain a baseline payoff of 0. Thus it is efficient to transport the fruit (we assume there are no other externalities) and transportation for any price between c_q and v_q is profitable for both F and T . If the fruit was properly refrigerated, F obtains high value v_H and T incurs high cost c_H . If the fruit is shipped without refrigeration, F obtains low value $v_L < v_H$ and T incurs lower cost $c_L < c_H$. We assume that $v_H - c_H > v_L - c_L > 0$, i.e., refrigeration is efficient as it results in higher total surplus.⁷

The fruit delivery and the payment from F to T are both verifiable.⁸ In case of a dispute, each party $i = \{F, T\}$ bears the cost λ_i of legal action, which does not depend on who initiated the action and who prevails. As mentioned earlier, the cost of enforcing agreements via the courts or arbitration can be very high compared to the potential gains from trade, and is practically infinite when the transacting parties are anonymous and the interactions short-term. We also assume that the courts are fair and able to correctly enforce the contract terms in full; introducing uncertainty in enforcement would not change the nature of our results although we would need to adjust for expectations and any risk aversion.

The resulting contracting game is sequential. After the parties agree on a price p , the *delivery game* is played where T transports the fruit, and F pays (or not) p upon delivery.⁹ We analyze subgame perfect equilibria in different versions of the contracting game. For intelligibility of our visual representation in figures focuses on the delivery game (i.e., subgame after contracting stage). Figure 1 shows the base delivery game.

In the base setting, the price agreed between F and T cannot depend on the quality of transportation as the latter is not verifiable by a third party, such as a court. In this case the dominant strategy for T is to provide low quality transportation, as refrigeration is costly. The price promised to T upon delivery needs to be at least λ_T , i.e., sufficiently large that it would be worthwhile for T to enforce the payment if F were to renege on paying. Thus low-quality transportation (without refrigeration) will be contracted for and take place if $\lambda_T \leq p$, which is possible only if $\lambda_T \leq v_L$. If the legal cost is larger than the value of the delivered good to F , there is no price F is willing to pay at which T would agree to deliver.¹⁰ The resulting equilibrium in the one-shot game is never efficient, as transportation either is low-quality or does not take place at all. This result is stated in the following Lemma.

⁷In our analysis, we focus on the relationship between T and F and whether trade takes place; thus we use *total surplus* to refer to the dyadic surplus of T and F . Specifically, we do not consider other agents in the economy such as customers of F , suppliers of T , or competitors to both companies.

⁸For instance, verifiable digital information about delivery via maritime shipping may be obtained via systems such as the Automatic Identification System, and commercial payment transactions leave a verifiable digital trail.

⁹For simplicity we assume that F would either pay or not pay; in real situations F could also pay with significant delays, which would also be costly for T and lead to qualitatively similar strategic outcomes.

¹⁰Gans (2021) obtains a similar result. In his model, the cost of providing verifiable evidence plays the same role in leading to failure of trade as the cost of legal action does in our setting.

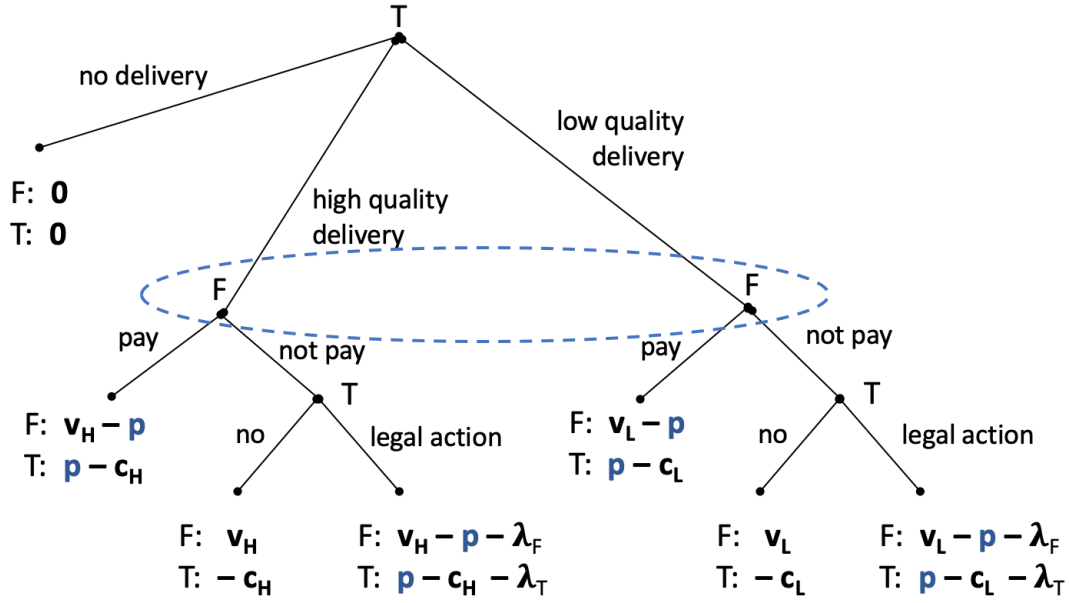


Figure 1: Base delivery game (without smart contract or additional sensors) in extensive form.

Lemma 1 *In the base contracting game (without smart contract or additional sensors), the equilibrium is never efficient.*

- For $\lambda_T \leq v_L$, low quality transportation is contracted and executed, at price $\max\{\lambda_T, c_L\} < p^0 < v_L$, yielding profits $\Pi_F^0 = v_L - p^0$ and $\Pi_T^0 = p^0 - c_L$. The gains from trade are $v_L - c_L > 0$.
- For $\lambda_T > v_L$, there is no contracting. Profits and gains from trade are 0.

Proof. All proofs are in the appendix.

Even though in equilibrium legal action is never taken, T 's cost of legal action λ_T crucially affects the equilibrium outcome as it determines whether the threat of legal action is credible or not. When the cost of legal action is high enough for T , specifically when $c_L < v_L < \lambda_T$, there will be no trade (i.e, agreement to transport the fruit) even though transporting would be efficient. Even when trade occurs in equilibrium, it is inefficient, as low quality transportation is contracted, even though high quality transportation would be more efficient.

In settings with repeated interaction over long periods these inefficiencies can sometimes be addressed with relational contracts or reputation mechanisms (Bakos and Dellarocas 2011). The literature on social contract enforcement points out that when these mechanisms prove inadequate, institutions may arise invoking a much stronger society-wide punishment for reneging on agreements, e.g., indefinite exclusion from the trade with whole group (Greif 1993, Tirole 1996).

Another possibility would be vertical integration, along the analyses of transaction cost theory (Williamson 1975) or Incomplete Contracts (Hart and Moore 1988, 1999).

These solutions, however, favor larger and older companies, while putting smaller and newer entrants at a disadvantage. Such imbalance creates barriers to entry, and eventually inefficiency in the market, as more efficient entrants without established reputation find it difficult to compete against less efficient but established incumbents. Proponents of smart contracts argue that smart contracts could address these inefficiencies without relying on relational contracts, reputation or vertical integration, and thus alleviate barriers to entry, leading to a more competitive landscape. We evaluate such claims in the following sections.

Posting a bond also provides a mechanism to avert reneging on agreed obligations, but typically either requires high cost mechanisms such as obtaining letters of credit and establishing arbitration to resolve disputes, or relies on platforms with trusted intermediaries such as eBay, PayPal, Amazon or Alibaba that escrow the payments and resolve disputes. These platform intermediaries also present a costly way to prevent reneging, both because of their transaction fees and the limitations of their dispute resolution processes, which typically makes them unsuitable for high value transactions.

4 Impact of introducing smart contracts

Even though refrigeration during transportation cannot be verified, both the fruit delivery and F 's payment to T are verifiable and are also likely to be captured by the existing technology infrastructure, e.g., as digital confirmation of delivering a shipping container or processing a bank transfer. Thus we assume that delivery by T and payment by F can be used as digital inputs to a smart contract without the need for additional digital inputs or authentication.¹¹ With a smart contract, payment can be executed automatically after delivery is established. This means that F cannot renege on payment, which truncates the strategy space, as is represented in Figure 2.¹² A smart contract protects T against being held up by F by automating payment upon delivery and this protection adds value when the legal cost T would incur in enforcing the contract is so high that it would prevent contracting. This leads to the following Lemma:

¹¹Certain digital inputs that can be obtained from the existing technology infrastructure will identify certain states and actions that are currently verifiable and thus can be incorporated as triggers for smart contracts. By contrast, for other states and actions verifiability would require the deployment, and possibly the development, of appropriate technology such as new IoT sensors, and these states and actions can only be incorporated in smart contracts once this technology is in place. Notably, sensors do not need to be perfect; they add value even if they provide a noisy signal, as long as that signal is informative.

¹²Smart contracts may be employed because they enable cost-effective automation; while this could change the payoffs in our game by creating additional surplus, we focus our analysis on the *commitment* aspect of smart contracts, which would change the nature of the game itself by preventing reneging.

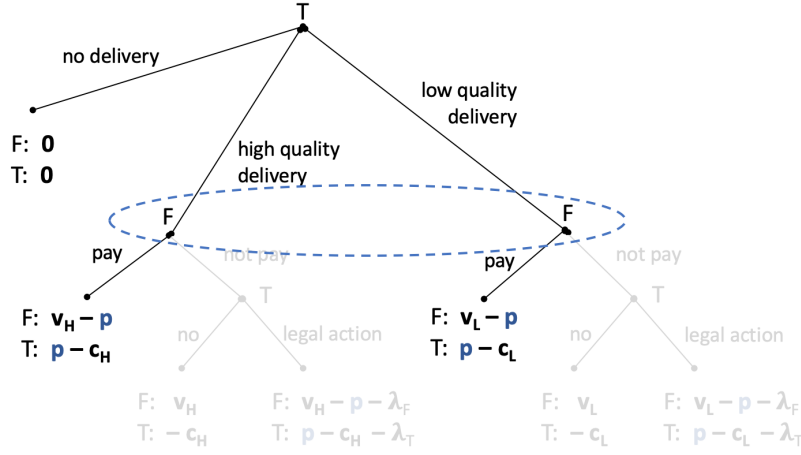


Figure 2: Delivery game with a smart contract conditioned on delivery (and no additional sensors). Strategies which are no longer available to F due automated execution are shaded.

Lemma 2 *In the contracting game with only smart contracts, low quality transportation is contracted and executed whenever $c_L < v_L$, at price $c_L < p^C < v_L$, yielding profits $\Pi_F^C = v_L - p^C$ and $\Pi_T^C = p^C - c_L$. The gains from trade are $v_L - c_L > 0$.*

Extending the contracting region over which trade takes place improves efficiency because for the parameter values that trade takes place the payoff of both firms increases. Thus, in this setting, smart contract technology improves efficiency by extending the contracting region. Smart contracts, however, do not improve the resulting trade outcomes, and thus contracting still falls short of achieving full efficiency.

5 Impact of adding connected sensors

Progress in sensor and connectivity technologies enables the development of reliable connected sensors with readings that can be viewed, recorded and trusted by the appropriate parties. Such sensors, for instance, can measure the temperature inside the shipping container, thus making possible to verify whether the fruit was properly refrigerated during transportation. This allows F (and third parties such as a court or an arbitrator) to distinguish between the two different states—refrigerated and not refrigerated transportation—and to set a different contract price for each, which we denote as p_H and p_L respectively. Figure 3 shows how this affects the contracting game.

Since $v_H - c_H > v_L - c_L$, a Pareto optimal contract will have p_L and p_H such that $v_H - p_H > v_L - p_L$ and $p_H - c_H > p_L - c_L$. Under such a contract, low quality transportation is never provided in the equilibrium. To ensure that F will not renege on its payment to T for high quality

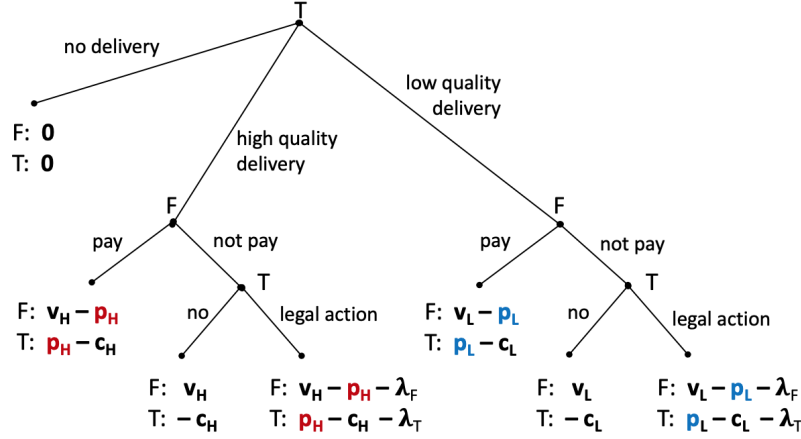


Figure 3: Delivery game with connected sensors verifying the quality of transportation (and no smart contracts).

transportation, it is necessary that $\lambda_T < p_H$, which is possible only if $\lambda_T < v_H$.

Note that for p_H and p_L such that $p_L > \lambda_T$ and $p_L - c_L > p_H - c_H$, T would provide low quality transportation. However, neither T or H would find it profit-maximizing to offer such prices, if they were proposing the contract terms. This leads to the following Lemma:

Lemma 3 *Adding connected sensors that verify the quality of transportation (but no smart contracts) leads to trade that is efficient when it takes place, but there is no trade when T faces high enough cost of legal action.*

- For $\lambda_T \leq v_H$, high quality transportation is contracted and performed, at price p_H^S where $\max\{\lambda_T, c_H\} < p_H^S < v_H$, yielding profits $\Pi_F^S = v_H - p_H^S$ and $\Pi_T = p_H^S - c_H$. The gains from trade are $v_H - c_H > 0$.
- For $\lambda_T > v_H$, there is no contracting. Payoffs and gains from trade are 0.

Compared to the base case, the added connected sensors improve both the region of contracting, and the efficiency of the contracted outcome when contracting takes place. Connected sensors do not achieve full efficiency, however, in the sense that certain efficient trades will not take place when T faces a high cost of legal action.

Compared to the employing smart contracts only, there is less expansion of the contracting region, but the efficiency of contracting is improved when contracting does take place. Connected sensors expand the contracting region to the interval (c_H, λ_T) , which is smaller than the (c_L, λ_T) interval by which the contracting region is extended when implementing smart contracts only. Whenever contracting takes place, however, connected sensors increase the gains from trade to $v_H - c_H$, while smart contracts offer only $v_L - c_L$.

6 Using smart contracts and connected sensors together

Introducing both a smart contract and new connected sensors that can verify the quality of transportation increases the state space over which F and T can contract, by allowing F and the courts to determine whether T provided refrigeration during transport, and limits strategy space by not allowing F to renege on payment. The game with both a smart contract and sensors is represented in Figure 4. As a result, contracting and trade take place when socially efficient, i.e., when $c_H < v_H$, at a price between these values, as described in the following Lemma.

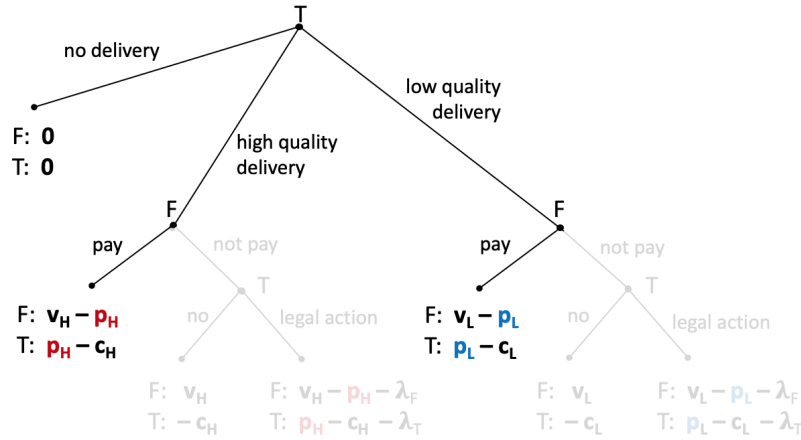


Figure 4: Delivery game with both connected sensors verifying the quality of transportation and a smart contract based on the quality of transportation.

Lemma 4 *In the contracting game with both connected sensors and smart contracts, contracting is fully efficient. High quality transportation is contracted for and executed whenever $c_H < v_H$ at price p_H^{C+S} where $c_H \leq p_H^{C+S} \leq v_H$, yielding profits $\Pi_F^{C+S} = v_H - p_H^{C+S}$ and $\Pi_T^{C+S} = p_H^{C+S} - c_H$. The gains from trade are $v_H - c_H > 0$.*

Lemmas 1–4 illustrate that the two technologies—connected sensors and smart contracts—differ in their capabilities and their resulting impact on the underlying strategy game. Specifically, **smart contracts restrict the strategy space by removing the ability to renege, while connected sensors allow to verifiably distinguish more states of nature**. Thus, in the game represented in Figure 4, smart contracts extend the contracting region, while connected sensors improve the efficiency of contracting when such contracting takes place.

This point is worth emphasizing because the benefits of smart contracts and connected sensors are often confounded in popular discourse. Sensors can be implemented without smart contracts, smart contracts can be based on existing digital inputs without the need for additional connected sensors, and each of these technologies has its own implementation cost. We distinguish and

characterize the implications of automated execution (which is at the core of smart contracts) and the more granular states of the world that can be identified by connected sensors, and how they each affect the scope and efficiency of contracting. Our analysis thus examines under what conditions it is most beneficial to pursue smart contracts, set up new sensors, or apply both technologies together.

Table 1 summarizes the different ways in which smart contracts and connected sensors improve efficiency in our setting. Smart contracts automate certain actions, and thus create commitment by limiting the strategy space. This results in increased efficiency by preventing hold-up and extending the contracting region. Smart contracts without new sensors extend the contracting region to $\lambda_T > v_L$, and with connected sensors that can show refrigeration extend it to $\lambda_T > v_H$. Connected sensors that provide verifiable proof of the quality of transportation extend the contracting space even in one-shot interactions and improve efficiency by ensuring that when trade occurs, it is for high quality transportation. Sensors alone, however, do not extend the contracting region when λ_T is high. Similarly, smart contracts alone do not enable contracting for high quality transportation.

		SENSORS	
		NO	YES
SMART CONTRACT	NO	(Lemma 1) low-quality or no contracting	(Lemma 3) high-quality or no contracting
	YES	(Lemma 2) low-quality	(Lemma 4) high-quality (efficient)

Table 1: Equilibrium properties under different technologies.

That means that each of these technologies may improve efficiency or provide no benefit, depending on the situation. There is also a range of parameters where a positive benefit can be obtained only when the two technologies are implemented together, which we explore in more detail in the next section.

7 Tradeoffs in Deployment of Smart Contracts and Connected Sensors

Improved connected sensors can create value with or without smart contracts, by enabling trades with high quality transportation and thus increasing total surplus. They may also enable certain trades (by extending the contracting region) even in the absence of smart contracts. Smart contracts create value by enabling trades that were prevented by fear of contractual renegeing. Thus, each technology is improving contracting outcome, but not uniformly. And also implementing both technologies does not necessarily provide compound benefit. In our setting, when smart contracts are implemented for trades that would be executed anyway, they do not improve the efficiency of these trades and thus they decrease total welfare if we take account of their implementation cost.¹³

The following Lemma states these results formally:

Lemma 5 *Both smart contracts and additional connected sensors improve contracting outcome, but not uniformly. The value added by sensors and smart contracts depends on λ_T as follows:*

- *Without smart contracts, sensors increase the gains from trade by $v_H - c_H - (v_L - c_L) > 0$ for $\lambda_T < v_L$, and they increase the gains from trade by $v_H - c_H$ for $\lambda_T \in (v_L, v_H)$.*
- *When added to smart contracts, sensors always increase the gains from trade by $v_H - c_H - (v_L - c_L)$.*
- *Without sensors, smart contracts increase the gains from trade by $v_L - c_L$ for $\lambda_T > v_L$.*
- *When added to sensors, smart contracts increase the gains from trade by $v_H - c_H$ for $\lambda_T > v_H$.*

The results of Lemma 5, which are illustrated in Figure 5, suggest that smart contracts may indeed help to facilitate small-value trades where the gains from trade would be smaller than legal costs, or enable transactions with small agents that would not be able to afford legal action, as frequently claimed in the popular literature.

IoT sensors and smart contracts each weakly increase the gains from trade and thus total welfare, which is shown in Figure 5 by the value added being always positive. Thus, if there would be no cost to implementing these technologies, it would be efficient to always introduce them. However, as we show below, it may not be worth introducing one or both if the cost is non-negligible. Moreover, while smart contracts and connected sensors may (weakly) increase the total gains to trade, it does not guarantee that both parties benefit from this increase. As we show in Section 8, it is possible that one of the parties may be strictly worse off from the introduction of these technologies.

¹³Automation provided by smart contracts can result in operational efficiency improvements, which we do not model in our setting, and which can offset their implementation cost and thus result in increased total surplus.

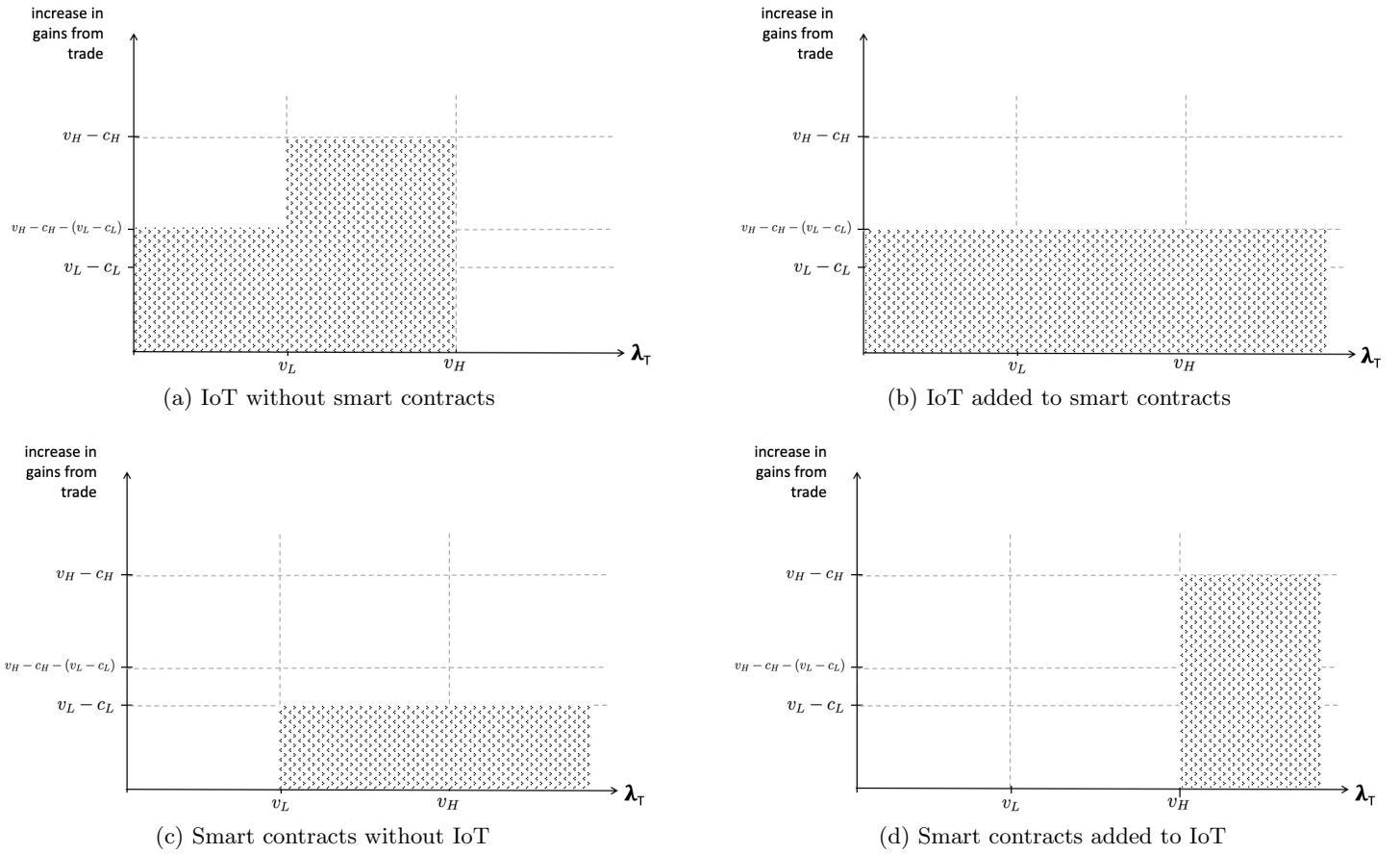


Figure 5: Value added of connected sensors (IoT) and smart contracts.

Propositions 1 and 2 follow directly from Lemma 5 as illustrated in Figure 5.

Proposition 1 *When $\lambda_T \in (v_L, v_H)$, smart contracts and additional connected sensors are substitutes, in the sense that depending on the parameter values it may be socially efficient to implement the sensors alone, or smart contracts alone, but it is never socially efficient to implement both technologies.*

Proposition 2 *When $\lambda_T > v_H$, smart contracts and additional connected sensors are complements, in the sense that the benefit of one technology is enhanced in the presence of the other. In this implementing case sensors are socially efficient only if they are efficiently implemented together with smart contracts.*

To assess which, if any, technology is socially beneficial to implement, we need to weigh the benefits of each technology against its cost of implementation. In the case of smart contracts, the primary cost of implementation would include the cost of linking the necessary digital inputs,

programming the algorithmic part, and ensuring execution of the contract itself as well as the actions prescribed by its execution.¹⁴ In the case of connected sensors, there may be significant development costs, in addition to the cost of deploying and operating the sensors.

We use $\kappa_C > 0$ and $\kappa_S > 0$ to denote the cost of implementing smart contracts and sensors, respectively. If both technologies are implemented, both costs are incurred. As shown in Figure 5, with the exception of adding sensors capable of verifying refrigeration to already implemented smart contracts, there are certain values of parameters in our setting where introducing a technology will provide no incremental benefit, and thus it may not be socially optimal to incur the cost of implementing one or both technologies even if their costs are low. Proposition 3 specifies the optimal implementation of technologies in terms of κ_C and κ_S .

Proposition 3 *Social benefit from implementing smart contracts and connected sensors depends on the cost of their implementation, κ_C and κ_S respectively, as well as T 's cost of legal action, λ_T , as follows:*

(1) *for $\lambda_T < v_L$, it is socially optimal to implement sensors if $\kappa_S \leq v_H - c_H - (v_L - c_L)$, and it is not beneficial to implement smart contracts, at any $\kappa_C > 0$;*

(2) *for $\lambda_T \in (v_L, v_H)$, it is socially optimal to implement*

- *sensors, but not smart contracts when*

$$\kappa_S \leq v_H - c_H \text{ and } \kappa_S - \kappa_C \leq v_H - c_H - (v_L - c_L),$$

- *smart contracts, but no sensors when*

$$\kappa_C \leq v_L - c_L \text{ and } \kappa_S - \kappa_C \geq v_H - c_H - (v_L - c_L);$$

(3) *for $\lambda_T > v_H$, it is socially optimal to implement*

- *sensors and smart contracts when*

$$\kappa_C + \kappa_S \leq v_H - c_H \text{ and } \kappa_S \leq v_H - c_H - (v_L - c_L),$$

- *smart contracts, but not sensors when*

$$\kappa_C \leq v_L - c_L \text{ and } \kappa_S \geq v_H - c_H - (v_L - c_L).$$

The results of Proposition 3 are illustrated in Figure 6.

As can be seen from Figure 6, when $\lambda_T > v_H$, i.e., when the cost to enforce a contract is very high, sensors improve the equilibrium only if they can be implemented together with smart

¹⁴Contract execution itself may require resources, such as gas on the Ethereum platform.

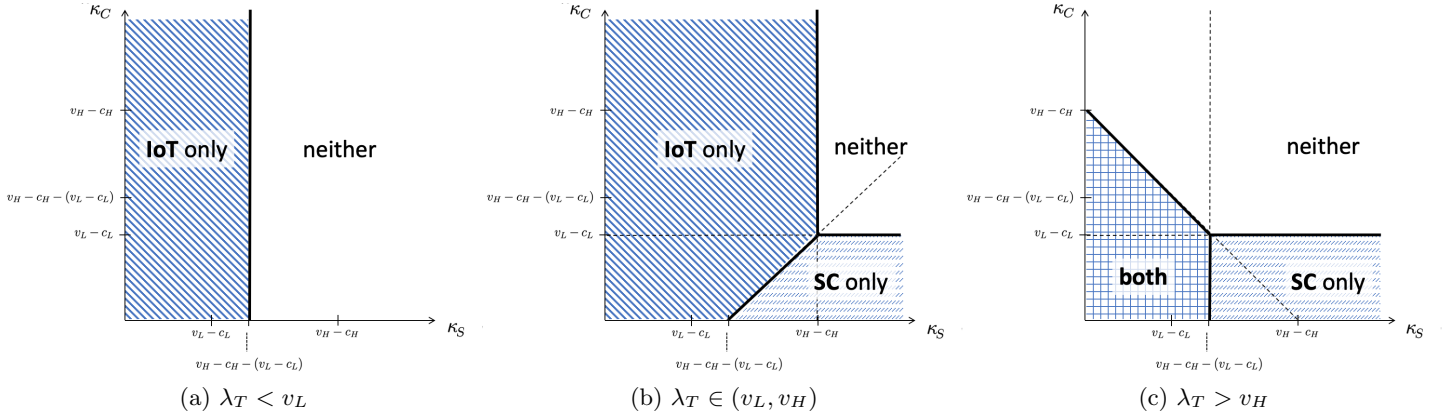


Figure 6: Socially optimal implementation of smart contracts and connected sensors, depending on κ_C , κ_S and λ_T .

contracts; this is because smart contracts are necessary to prevent F from reneging on the agreed payment. Thus, for $\lambda_T > v_H$, when $v_H - c_H > \kappa_C + \kappa_S$ but $v_L - c_L < \kappa_C$, it is socially optimal to implement both technologies together while it is suboptimal to implement either one alone.

When κ_S , the cost to implement connected sensors, is less than $v_H - c_H - (v_L - c_L)$, then it is socially efficient to employ sensors to guarantee high quality transportation. In order for T to provide the required refrigeration, T must receive a price high enough to protect from being held up, and that means a price of at least λ_T , so that T would enforce the contract if F reneges on payment. In that case if $\lambda_T < v_H$, then implementing smart contracts reduces total surplus; smart contract technology will not change the gains from trade, and thus it will not offset the cost of its implementation. Smart contracts may increase the surplus captured by F , however, and thus F may favor their adoption; as discussed in the next section, this could lead to socially inefficient adoption of smart contracts.

We see that λ_T , T 's cost of enforcing a contract, is a key parameter in determining the social benefit from adopting smart contracts, connected sensors, or both. For medium values of λ_T compared to the potential benefits from trade it may be appropriate to employ one or the other, while for high values of λ_T it may be appropriate to employ both. For low λ_T , smart contracts are not beneficial to implement, as conventional contracts will sufficiently protect T from being held up, unless smart contracts can be justified solely on their *benefits from process automation*.

8 Total surplus vs incentives for adoption

The above results show that both smart contracts and connected sensors will increase gains from trade for certain parameter values, and thus in these cases it is socially optimal to implement one or both if the implementation cost is sufficiently low. Similarly, there are parameter values where smart contracts and connected sensors add little or no value, and thus may not offset the cost of their implementation, making it socially optimal not to adopt them. However, T and F may differ in their private incentives, as adoption may improve payoff for one party, but decrease payoff for the other. Furthermore, T and F 's private incentives may diverge from what is socially optimal.

To explore these questions, we use $\gamma \in (0, 1)$ to denote the bargaining power that T has against F , meaning that T can capture fraction γ of any incremental surplus from trade. We treat γ as exogenously determined by factors not captured in our setting, such as the parties' available alternatives. For simplicity we assume that γ does not depend on whether smart contracts or connected sensors are adopted.¹⁵ This formulation allows us to compare the relative benefit of adopting each technology for F and T and the resulting incentives to adopt, and also see if it is possible for one of them to be worse off as a result of adoption.

8.1 It is possible to over-provide smart contracts

In the base scenario without smart contracts or sensors that can verify refrigeration, trade can take place only if $p \geq \lambda_T$, which may allow T to capture a larger share of the surplus than is warranted based on bargaining power; essentially F may need to yield some surplus to T so that T will be protected against reneging on payment. As a result, when T 's bargaining power is low, F may have an incentive to introduce smart contracts even when this would decrease social surplus.

When trade takes place without smart contracts or sensors only low-quality (unrefrigerated) transportation is provided, resulting in surplus of $v_L - c_L$ and in the absence of other constraints, the agreed price p would be $\gamma(v_L - c_L) + c_L$. However, a minimum price of $p > \lambda_T$ is required to prevent F from reneging on payment, and the resulting price without smart contracts or sensors is $p^0 = \max\{\gamma(v_L - c_L) + c_L, \lambda_T\}$, with trade possible when $\lambda_T < v_L$. When γ is so low that $\gamma(v_L - c_L) + c_L < \lambda_T$, T will be offered price λ_T , higher than what it could obtain based on its bargaining power, as a price below λ_T would leave T exposed to being held up by F .

Introducing smart contracts removes the constraint $p > \lambda_T$ and expands the contracting space, allowing trades for $v_L < \lambda_T$ as long as $v_L > (c_L + \kappa_C)$. When smart contracts enable such new trades, both parties gain and this leads to higher social surplus. When $(\gamma(v_L - c_L) + c_L) < \lambda_T \leq v_L$,

¹⁵It is possible that adoption of smart contracts and/or connected sensors can affect the relative bargaining power of the trading parties, and this could be an interesting topic for future research.

however, without smart contracts trade takes place at price $p^0 = \lambda_T$ while with smart contracts the price is $p^C = \gamma(v_L - c_L) + c_L < p^0$. Implementing smart contracts for this parameter range shifts the value captured from T to F , and since the gains from trade $v_L - c_L$ do not change and the implementation of smart contracts is costly ($\kappa_C > 0$), the total surplus $v_L - (c_L + \kappa_C)$ is lower.

Thus it is possible to have a divergence between private and social incentives in our setting. If F , as the party that makes payment and has the stronger bargaining position, can unilaterally prescribe the use of smart contracts, this could lead to an over-provision of this technology. In fact, F would be willing to spend up to $\lambda_T - (\gamma(v_L - c_L) + c_L)$ to introduce smart contracts and thus capture more surplus, even though it would be socially inefficient as the cost of implementation would be incurred without increasing the gains from trade.

Corollary 1 *In a setting where only smart contracts are available, for some parameter values F may benefit from introducing smart contracts that decrease total surplus. Specifically:*

- (1) *For $\lambda_T \in (\gamma(v_L - c_L) + c_L, v_L]$, it is individually beneficial for F to adopt smart contracts, as $p^C < p^0$, and thus $\Pi_F^C > \Pi_F^0$, even though it is socially disadvantageous to adopt the technology, as $\Pi_F^C + \Pi_T^C - \kappa_C < \Pi_F^0 + \Pi_T^0$.*
- (2) *For $\lambda_T > v_L$, it is individually beneficial to adopt smart contracts whenever it is socially optimal to do so, as for any $\gamma \in (0, 1)$, $\Pi_i^C > \Pi^0 = 0$ for both $i = \{F, T\}$.*
- (3) *For $\lambda_T < \gamma(v_L - c_L) + c_L$, incentives to adopt are also aligned with social welfare, as it wouldn't be socially optimal to implement smart contract and neither party has incentive to do so, since $p^C = p^0$ and $\Pi_i^C = \Pi^0$ for both $i = \{F, T\}$.*

8.2 Sensors can increase total surplus but make T worse off

When T 's bargaining power is low enough, T may also be made worse off by the introduction of sensors that can verify refrigeration. This is because as explained in section 8.1, in the base case without sensors or smart contracts trade can take place only if $p \geq \lambda_T$ so that T will be protected from F reneging on payment, and will be for non-refrigerated transportation. Introducing sensors that can verify refrigeration may shift the equilibrium to refrigerated transportation, which will increase T 's cost by $c_H - c_L$. Unless the surplus from refrigerated transportation is so much higher that T 's share is larger than $\lambda_T + (c_H - c_L)$ to compensate for the increased transportation cost, T will be worse off.

For instance, when $\gamma(v_H - c_H) + c_L < \lambda_T < v_L$, i.e., $\gamma < \frac{\lambda_T - c_L}{v_H - c_H} < \frac{v_L - c_L}{v_H - c_H}$, in the base case trade will take place at price $p^0 = \max\{\gamma(v_L - c_L) + c_L, \lambda_T\}$. Since $\gamma(v_L - c_L) + c_L < \gamma(v_H - c_H) + c_L < \lambda_T$,

the price will be $p^0 = \lambda_T$ and T 's profit will be $\Pi_T^0 = \lambda_T - c_L$. Adding sensors that can verify whether transportation was refrigerated allows to specify different prices for high and low quality delivery. When $v_H - p_H > v_L - p_L$ and $p_H - c_H > p_L - c_L$, high quality (i.e., refrigerated) transportation is provided at equilibrium, resulting in total surplus $v_H - c_H$, price $\max\{\gamma(v_H - c_H) + c_H, \lambda_T\}$, and T 's profit $\Pi_T^S = \max\{\gamma(v_H - c_H), \lambda_T - c_H\}$. Since $\lambda_T > \gamma(v_H - c_H) + c_L$ and $c_H > c_L$, $\Pi_T^S < \Pi_T^0$. Thus, T would be worse off after the implementation of sensors, even though the total surplus would be higher, as T would bear most of the cost of providing refrigeration. The following corollary provides a more detailed characterization of parameter values leading to this outcome:

Corollary 2 *In a setting where sensors capable to verify refrigeration during transportation but no smart contracts can be introduced and $\gamma < \frac{v_L - c_L}{v_H - c_H}$, for some parameter values T is worse off if sensors are implemented even though this increases the total surplus. Specifically:*

- (1) *For $\lambda_T \in (\gamma(v_H - c_H) + c_L, v_L]$, adoption of sensors makes T worse off, as $\Pi_T^S < \Pi_T^0$, even though it increases the gains from trade $\Pi_F^S + \Pi_T^S > \Pi_F^0 + \Pi_T^0$, and also increases the total surplus when $\Pi_F^S + \Pi_T^S - \kappa_S > \Pi_F^0 + \Pi_T^0$.*
- (2) *For $\lambda_T < \gamma(v_H - c_H) + c_L$ and for $\lambda_T \in (v_L, v_H)$, if implementing sensors increases total surplus, there exists a cost sharing rule such that it is individually beneficial for both F and T to implement the sensors, in the sense that for any $\gamma \in (0, 1)$, $\Pi_i^S \geq \Pi_i^0 = 0$ for both $i = \{F, T\}$.*
- (3) *For $\lambda_T > v_H$, there will be no trade with or without the sensors, neither party has incentive to introduce them, all profits are zero, and in that sense individual incentives are aligned with social incentives.*

It should be noted that in the first case of the above corollary, adding smart contracts in addition to the sensors does not alleviate the fact that T is worse off after implementing the sensors, even though this is socially efficient. This is because for this range of parameters, $\Pi_T^{C+S} \leq \Pi_T^S < \Pi_T^0$, in other words T may be even worse off if smart contracts are also introduced.

9 The Role of Blockchains

Blockchain technology is central in our setting and analysis in three main aspects:

First, a well-implemented blockchain technology gives assurance of irreversibility for actions executed by smart contracts, such as payment upon delivery; for instance, this irreversibility prevents F from reneging on payment according to the agreement with T . Before blockchains, the value of smart contracts was automation rather than commitment. For instance, automated payment

with a credit card may offer convenience, but it can be reversed by the credit card network if it is challenged as unauthorized or fraudulent. In our setting, an automated but reversible payment would not alleviate T 's concerns when λ_T is high.

While for the clarity of our analysis it was appropriate to assume that transactions triggered by smart contracts are fully irreversible, different types of blockchain platforms provide different levels of assurance against reversing a transaction. In permissioned blockchains a payment to T *can* be reversed by agreement of a large enough number of validators, but such reversals are very rare and would likely require an extraordinary level of effort and supporting evidence on F 's part. As long as the blockchain operator and validators can be trusted not to compromise the platform, T is well protected. Similarly in well-functioning permissionless blockchains transactions are also practically irreversible, as reversal would require either a successful attack or a level of coordination among miners similar to what is involved in protocol changes, either of which would be infeasible for F to accomplish.

Second, blockchain technology guarantees that information provided from sensors and oracles has not been tampered with after it is recorded on the blockchain. Modern sensors can use digital signatures to assure the integrity of their data and can be designed to prevent or at least detect tampering with their operation. A concern when deployed is to prevent other forms of manipulation, such as keeping the sensor temperature low without properly refrigerating the shipment itself. Oracles can rely on data from sensors, but may also rely on reported data—e.g., utilizing APIs from websites. Digital signatures can protect the integrity of this data during transmission to the oracle and then recording to the blockchain, but the concern remains that the sensors, websites or other data sources can be manipulated.

The capabilities of connected sensors are continuously increasing as their hardware and software improves, and so does the network infrastructure that enables their connectivity as demonstrated by the rapidly increasing number of devices and expanding functionality of the Internet of Things. This increase in capability, combined with the authentication and integrity protection provided by digital signatures and recording on the blockchain, can allow to verifiably identify the occurrence of states of the world at a higher level of detail, and use smart contracts to trigger appropriate corresponding actions.

Third, there is a substantial recent growth in decentralized business models based on short-term interactions among anonymous or pseudonymous parties that are typically implemented on permissionless blockchains such as Ethereum. In such interactions the mechanisms typically used to enforce agreements in business settings, such as the court system or alternatives like reputation or relational contracts, are not feasible and the cost λ of enforcing agreements, which is a key parameter in our analysis, is practically infinite. As a result, smart contracts and trusted oracles

or sensors that can provide the necessary digital inputs are key in enabling these business models; for instance, decentralized finance (DeFi) depends on transactions among parties that do not know or trust each other and is made possible only because all aspects of such transactions can be implemented by smart contracts that are triggered based on information from trusted sources—trust is in the technology rather than in the institutions.

10 Discussion and Concluding Remarks

Smart contracts prescribe actions that are algorithmically executed upon satisfaction of agreed upon conditions, as in a brokerage limit order. While they have existed long before blockchain technology, blockchains provide an infrastructure for the recording and execution of smart contracts, certify the occurrence of contracted states of the world, and provide irreversible execution of certain actions such as payments “on the blockchain.” Blockchain-based smart contracts have been a key enabler for business models involving short-term interactions among potentially anonymous parties that do not know and do not trust each other, and thus cannot rely on the legal system for enforcement, such as in Decentralized Finance (DeFi).

Smart contracts require digital inputs to determine when the conditions for the execution of their prescribed actions should take place, which typically are provided on the blockchain by connected sensors or blockchain oracles. Predictions for the coming prevalence of smart contracts are thus predicated on the availability of connected sensors such as those that are part of the Internet of Things. Progress in sensor technology enables them to identify more states of the world with high accuracy and at a fine-grained level of discrimination, and frequently makes these states “verifiable” as the sensors provide evidence that can be shown to a court or an arbitrator in the event of a dispute. The two technologies are thus intricately linked but have distinct implications.

We developed a model that distinguishes the impact of smart contracts and connected sensors and used it to analyze their impacts on contracting outcomes. We considered a setting with two agents, T and F , where T provides transportation for F and can exert a costly effort that affects the value of the delivered good. We analyzed four possible regimes: (a) conventional contracting; (b) contracting with increased information from connected sensors; (c) smart contracts that automate certain actions; and (d) the combination of smart contracts and connected sensors. Our analysis applies to more general settings as long as the relevant parameters are codifiable, i.e., the effort can be measured directly or indirectly with sensors and the subsequent actions can be algorithmically described so they can be encoded in a smart contract.

We explored the impact of introducing smart contracts by considering automated and irrevocable payment upon delivery by T . For this to work, confirmation of the agreed condition (in this

case delivery) must be digitally available on the blockchain so that it can trigger the corresponding action. In practice this may be achieved either via direct blockchain verification, such as a digital token representing proof of delivery, via an appropriately verified sensor reading, such as scanning a tamper proof tag on a shipping container, or via a trusted oracle such as a system that obtains verified information from port authority or customs records, and provides a corresponding indication of delivery on the blockchain.¹⁶

We explored the impact of connected sensors by considering implementation of a sensor that can verifiably determine and communicate on the blockchain whether proper refrigeration was provided during transportation; this could be accomplished, for instance, by a sensor that indicates whether transport temperature was kept in the appropriate range, is able to detect tampering with either the container or the sensor itself, and can communicate its readings to the blockchain either directly or via a trusted oracle. Implementing this type of sensor in our setting would make refrigeration during transport verifiable, as the readings of the sensor can be provided to a third party such as a court or an arbitrator, and would also allow automated payment upon delivery for an amount that depends on whether refrigeration was provided.

We found that connected sensors and smart contracts have different impacts on contracting outcomes and efficiency, and thus should not be confounded. *Connected sensors increase verifiable information and thus expand the strategy space* by allowing payoffs to depend on actions and outcomes not previously verifiable. This can lead to more efficient trades, but typically does not achieve full efficiency, especially in settings where enforcement costs are high. By contrast, *smart contracts automate execution and thus restrict the strategy space* of the parties to the contract, typically by eliminating actions like reneging or hold-up, and thus allow commitments that previously would be expensive to enforce. This typically increases the contracting region where trade occurs, and thus increases efficiency. Finally, when applied together, smart contracts and connected sensors enable all efficient trades, including certain trades that neither technology can enable individually.

Our analysis distinguished the technologies of smart contracts and connected sensors in terms of their implications for contracting and economic efficiency. Predictions of the increasing prevalence of smart contracts often fail to make this distinction and incorrectly attribute the benefit from the digital inputs that provide information about the occurring states of the world, to the smart contracts that use them. This can bias theoretical analysis of the role of smart contracts, and promote inefficient implementation of technology in practice.

Technological progress in connected devices has meant that the Internet of Things also includes connected “actuators” that allow the triggering of *actions*, such as remotely disabling the car in

¹⁶Within the scope of this paper, blockchain oracles have the same implications as connected sensors, so everything we find about sensors also applies to blockchain oracles.

Szabo's smart lien example. Like sensors, the end result of such actuators is to enable or facilitate new actions, and therefore extend the strategy space of the contracting parties; thus they have similar implications in terms of their economic analysis, and our analysis of connected sensors can be thought as also encompassing connected actuators.

Blockchain platforms broaden the scope and applicability of smart contracts by providing an infrastructure for their recording and execution, by helping certify the occurrence of contracted states of the world, and by enabling execution of certain actions "on the blockchain," such as a cryptocurrency transaction. Furthermore, smart contracts in shipping industry platforms such as TradeLens (provided by IBM Blockchain Solutions in cooperation with Maersk) are largely enabled because of developments in sensor technology that allow information to be collected and provided as an input to these contracts without the need for a human to observe and report. Progress in connected sensor technology allows to incorporate them in traded goods, such as the shipping container in our fruit transportation example. These sensors provide additional information such as the location, condition and use of the good and therefore enable contracting based on more contingencies.¹⁷

The driving force of our results is the parameter λ_T , the cost of *ex-post* enforcing an agreement if F reneges. Frequently this cost is not considered in the contracting literature, which typically is concerned with uncertainty, and the conditions needed for the parties to be willing to enter into an agreement, and assumes that its terms will be fulfilled (or later renegotiated). Transaction cost theory and incomplete contracts theory do consider enforcement costs as an important reason why certain relationships are subject to hold-ups and certain actions are non-contractible. They use vertical integration and the ownership of assets as mechanisms to address these situations (e.g., Williamson 1975, Grossman and Hart 1986, Hart and Moore 1988).

We made certain assumptions that may limit our analysis. We considered a simplified setting, with only one strategic action for the participants: for T whether to refrigerate and for F whether to make the agreed payment. In order to focus on the impact of smart contracts and connected sensors, we assumed that while these technologies will be costly to implement, their implementation will succeed. Furthermore we assumed that once implemented, both smart contracts and connected sensors will operate perfectly, and thus we abstract from the risk of incorrect sensor readings or coding errors in smart contracts. In considering whether T would choose legal action at cost λ_T if F reneged on payment, we assume that the outcome would be in T 's favor and restore payment; thus we abstract from the possibility of an incorrect or unenforceable judgement. Accounting for these possibilities would complicate our analysis but would not change the qualitative nature of

¹⁷Consistent with this, we have been told that a large number if not a majority of developers at IBM Blockchain are working on creating and patenting sensors.

our results; it may be an interesting area for future research, however, to gain further insights by exploring analysis that focuses on these considerations.

We conclude by pointing out that repeated relationships (i.e., relational contracts) and reputation can also prevent reneging on contractual obligations and thus can mitigate the cost of contract enforcement; reliance on these mechanisms, however, creates barriers to entry for new market participants. A central premise of smart contracts (and more generally blockchain technology) has been that they would “democratize the marketplace,” countering the advantage that established large players enjoy even if they do not offer a better product. In our analysis, we examine whether smart contracts indeed allow for this premise to be realized; we find that while smart contracts can help address enforcement costs, in many cases the ability to deliver on this premise (and promise) depends on the simultaneous deployment of appropriate connected sensors.

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Appendix: Proofs of Lemmas and Propositions

Proof of Lemma 1

Proof. We solve the game by backward induction. Whether after low- or high-quality delivery, in the last decision node, T takes a legal action if and only if $p^0 > \lambda_T$. If $p^0 \leq \lambda_T$, T does not take a legal action.

Thus, if $0 < p^0 \leq \lambda_T$, F will prefer not to pay, independently of the quality of delivery, because paying yields $v_i - p$ and not paying $v_i > v_i - p$ (this is because there will be no court action). At the previous decision node, T decides not to deliver at all, as delivery yields $-c_i < 0$. Moreover, at $p \leq 0$, T will not deliver as well. Thus, no trade will occur when $p^0 \leq \lambda_T$.

If $p^0 > \lambda_T$, F prefers to pay, independently of the quality of delivery, and receive $v_i - p^0 > v_i - p^0 - \lambda_F$. Since the price paid is the same independently of the quality of delivery (as the quality is not verifiable), if T delivers, it is a low-quality delivery, as $p^0 - c_L > p^0 - c_H$. And T will not deliver at all when $p^0 \leq c_L$.

Therefore, there is no trade if the contract sets $p^0 \leq c_L$ or $p^0 \leq \lambda_T$. With no trade both parties obtain 0.

For $p^0 > \max\{\lambda_T, c_L\}$, T would take legal action, if F doesn't pay, whether provided high-or low-quality delivery. Given that, F will pay p^0 for high- or low-quality delivery. Given that, T provides low-quality delivery. T obtains a payoff of $p^0 - c_L$, and F obtains $v_L - p^0$. However, if $p^0 > v_L$, F 's payoff would be negative, and thus, F will not agree to a contract with $p^0 > v_L$.

In our environment, $v_L > c_L$. Thus, if $v_L > \lambda_T$, there exist p^0 such that $\max\{\lambda_T, c_L\} < p^0 < v_L$, i.e., both F and T find it beneficial to contract. Then contracting yields profits $\Pi_F^0 = v_L - p^0 > 0$, $\Pi_T^0 = p^0 - c_L > 0$, and the gains from trade are $v_L - c_L > 0$.

$v_L < \lambda_T$, there does not exist p^0 at which both parties find it beneficial to contract. With no contracting, the profits and gains from trade are 0. ■

Proof of Lemma 2

Proof. Given a contract setting price p^C , the fruit delivery game reduces to T choice between high quality delivery, low quality delivery and no delivery. High quality delivery is dominated by low quality delivery, as for any p^C , $p^C - c_H < p^C - c_L$. T chooses low-quality delivery when $p^C - c_L > 0$. F contracts if $v_L - p^C > 0$. Since $v_L < c_L$, there always exists $c_L < p^C < v_L$, i.e., both F and T find it beneficial to contract. Then contracting yields profits $\Pi_F^C = v_L - p^C > 0$, $\Pi_T^C = p^C - c_L > 0$,

and the gains from trade are $v_L - c_L > 0$. ■

Proof of Lemma 4

Proof. Given the contract setting prices p_L^{C+S} and p_H^{C+S} , the fruit delivery reduces to T choosing between the two types of delivery or no delivery at all. If $p_H^{C+S} - c_H > \max\{p_L^{C+S} - c_L, 0\}$, then T chooses high-quality delivery. F 's payoff then is $v_H - p_H^{C+S}$. There always exists $c_H < p_H^{C+S} < v_L$.

Moreover, since $v_L - c_L < v_H - c_H$, a contract that would yield low-quality delivery in equilibrium (i.e., $p_L^{C+S} - c_L > \max\{p_H^{C+S} - c_H, 0\}$ and $p_L^{C+S} < v_L$) is always dominated by a contract that would a contract that yields high-quality delivery in equilibrium. To see that, consider a contract with $p_L^{C+S}, \tilde{p}_H^{C+S}$ such that $v_L - c_L + c_H < \tilde{p}_H^{C+S} < v_H$. We know that such $\tilde{p}_H^{C+S}$ exists, because $v_L - c_L < v_H - c_H$. Then, $p_H^{C+S} - c_H > p_L^{C+S} - c_L > 0$ and $v_L - p_H^{C+S} > v_L - p_L^{C+S} > 0$, i.e., a contract yielding high-quality delivery in equilibrium Pareto dominates the contract that yields low-quality delivery in the equilibrium. Thus, optimal contracting always yields high-quality delivery, with profits $\Pi_F^{C+S} = v_H - p_H^{C+S} > 0$, $\Pi_T^{C+S} = p_H^{C+S} - c_H$, and the gains from trade are $v_H - c_H > 0$. ■

Proof of Lemma 5

Proof. Follows directly from the results on gains from trade in Lemmas 1–4. ■

Proof of Proposition 1

Proof. From Lemma 5, for $\lambda_T \in (v_L, v_H)$, and given κ_C and κ_S :

- adopting smart contracts alone increases social benefit by $v_L - c_L - \kappa_C$;
- adopting sensors alone increase social benefit by $v_H - c_H - \kappa_S$;
- and adopting both increases the gains from trade by $v_H - c_H - \kappa_C - \kappa_S$.

If adopting only one technology, adopting only smart contracts is optimal when $v_L - c_L - \kappa_C > \max\{v_H - c_H - \kappa_S, 0\}$; adopting only sensors is optimal when $v_H - c_H - \kappa_S > \max\{v_L - c_L - \kappa_C, 0\}$; adopting neither is optimal when $\max\{v_L - c_L - \kappa_C, v_H - c_H - \kappa_S\} < 0$.

Adopting both is always dominated by implementing at most one technology. To see that, suppose first that $v_L - c_L - \kappa_C > v_H - c_H - \kappa_S$, i.e., implementing only smart contracts is more

beneficial than implementing only sensors. Then $v_H - c_H - \kappa_C - \kappa_S < v_L - c_L - \kappa_C$, i.e., implementing only smart contracts yields higher benefit than implementing both technologies. Next, suppose that $v_H - c_H - \kappa_S > v_L - c_L - \kappa_C$, but then $v_H - c_H - \kappa_S > v_H - c_H - \kappa_C - \kappa_S$ indicates that implementing only sensors dominates implementing both technologies. ■

Proof of Proposition 2

Proof. From Lemma 5, for $\lambda_T > v_H$, and given κ_C and κ_S :

- adopting smart contracts alone increases social benefit by $v_L - c_L - \kappa_C$;
- adopting sensors alone increase social benefit by 0;
- and adopting both increases the gains from trade by $v_H - c_H - \kappa_C - \kappa_S$.

Note that even if $v_H - c_H - \kappa_S > 0$, implementing sensors alone brings no benefit. If $v_L - c_L - \kappa_C > 0$, then it will be efficient to implement smart contracts.

Moreover, if $v_H - c_H - \kappa_C - \kappa_S > 0$, it may be beneficial to implement both technologies, even if implementing smart contract alone would not be beneficial, i.e., $v_L - c_L - \kappa_C < 0$. Thus, the benefit of smart contracts is enhanced by implementation of sensors, by more than the benefit of sensors only. And conversely, the benefit of sensors is enhanced by the implementation of smart contracts. ■

Proof of Proposition 3

Proof. Follows directly from Lemma 5, by the same arguments as in the proofs of Propositions 1 and 2. ■

Proof of Corollary 1

Proof. Follows directly from Lemmas 6 and 7 as well as calculations in the proof of Corollary 3. Proofs of Lemmas 6 and 7 follow from the definition of γ and Lemmas 1 and 2. ■

Lemma 6 *Suppose that T has the bargaining power γ . Then the equilibrium of the benchmark contracting game (without smart contract or additional sensors), depends on λ in the following way:*

- (1) For $\lambda_T < \gamma(v_L - c_L) + c_L$, trade occurs at $p^0 = \gamma(v_L - c_L) + c_L$, yielding $\Pi_T^0 = \gamma(v_L - c_L)$ and $\Pi_F^0 = (1 - \gamma)(v_L - c_L)$.
- (2) For $\lambda_T \in (\gamma(v_L - c_L) + c_L, v_L)$, trade occurs at $p^0 = \lambda_T$, yielding $\Pi_T^0 = \lambda_T - c_L > \gamma(v_L - c_L)$ and $\Pi_F^0 = v_L - \lambda_T < (1 - \gamma)(v_L - c_L)$.
- (3) For $\lambda_T > v_L$, trade does not occur, and thus $\Pi_T^0 = \Pi_F^0 = 0$.

Lemma 7 Suppose that T has the bargaining power γ . Then in the equilibrium of the contracting game with smart contracts only (without additional sensors) trade occurs at $p^C = \gamma(v_L - c_L) + c_L$, yielding $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$, for any λ_T .

Proof of Corollary 2

Proof. Follows directly from Lemmas 6 and 8 as well as calculations in the proof of Corollary 3. Proof of Lemma 8 follows directly from the definition of γ and Lemma 3. ■

Lemma 8 Suppose that T has the bargaining power γ . Then the equilibrium of the the contracting game with additional connected sensors only (without smart contract), depends on λ in the following way:

- (1) For $\lambda_T < \gamma(v_H - c_H) + c_H$, trade occurs at $p_H^S = \gamma(v_H - c_H) + c_H$, yielding $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$.
- (2) For $\lambda_T \in (\gamma(v_H - c_H) + c_H, v_H)$, trade occurs at $p_H^S = \lambda_T$, yielding $\Pi_T^S = \lambda_T - c_H > \gamma(v_H - c_H)$ and $\Pi_F^S = v_H - \lambda_T < (1 - \gamma)(v_H - c_H)$.
- (3) For $\lambda_T > v_H$, trade does not occur, and thus $\Pi_T^S = \Pi_F^S = 0$.

Proof of Corollary 3

Assume $\gamma < \frac{v_L - c_L}{v_H - c_H}$. Additionally assume $\gamma(v_H - c_H) + c_H < v_L$. Then consider following intervals

- $\lambda_T < \gamma(v_L - c_L) + c_L$
 $\Pi_T^0 = \gamma(v_L - c_L)$ and $\Pi_F^0 = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
 Thus, $\Pi_i^0 = \Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$.

- $\lambda_T \in (\gamma(v_L - c_L) + c_L, \gamma(v_H - c_H) + c_L)$
 $\Pi_T^0 = \lambda_T - c_L > \gamma(v_L - c_L)$ and $\Pi_F^0 = v_L - \lambda_T < (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
Thus, $\Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_F^0 < \Pi_F^C$ and $\Pi_T^C < \Pi_T^0 < \Pi_T^S$, but $\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^C$, so the differences in payoffs due to SC are only distributional.
- $\lambda_T \in (\gamma(v_H - c_H) + c_L, \gamma(v_H - c_H) + c_H)$
 $\Pi_T^0 = \lambda_T - c_L > \gamma(v_L - c_L)$ and $\Pi_F^0 = v_L - \lambda_T < (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
Thus, $\Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_F^0 < \Pi_F^C$ (so that $\Pi_F^0 < \Pi_F^C < \Pi_F^S = \Pi_F^{C+S}$) and $\Pi_T^0 > \Pi_T^S$ (so that $\Pi_T^C < \Pi_T^S = \Pi_T^{C+S} < \Pi_T^0$). Yet, $\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$.
- $\lambda_T \in (\gamma(v_H - c_H) + c_H, v_L)$
 $\Pi_T^0 = \lambda_T - c_L > \gamma(v_L - c_L)$ and $\Pi_F^0 = v_L - \lambda_T < (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \lambda_T - c_H > \gamma(v_H - c_H)$ and $\Pi_F^S = v_H - \lambda_T < (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
Thus, $\Pi_i^C < \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_T^C < \Pi_T^{C+S} < \Pi_T^S < \Pi_T^0$ and $\Pi_F^0 < \Pi_F^C < \Pi_F^S < \Pi_F^{C+S}$. (The inequality $\Pi_F^C < \Pi_F^S$ follows because in this interval, $\lambda_T < v_L < v_H + (1 - \gamma)(v_L - c_L)$.) Yet, $\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$.
- $\lambda_T \in (v_L, v_H)$
 $\Pi_T^0 = \Pi_F^0 = 0$;
 $\Pi_T^S = \lambda_T - c_H > \gamma(v_H - c_H)$ and $\Pi_F^S = v_H - \lambda_T < (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
Thus, $\Pi_i^0 < \Pi_i^C < \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_T^S > \Pi_T^{C+S}$ and $\Pi_F^0 < \Pi_F^S < \Pi_F^{C+S}$, but $\Pi_F^S > \Pi_F^C$. Also $\sum_{i=T,F} \Pi_i^0 < \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$.
- $\lambda_T > v_H$
 $\Pi_T^0 = \Pi_F^0 = 0$;
 $\Pi_T^S = \Pi_F^S = 0$;

$$\begin{aligned}\Pi_T^C &= \gamma(v_L - c_L) \text{ and } \Pi_F^C = (1 - \gamma)(v_L - c_L); \\ \Pi_T^{C+S} &= \gamma(v_H - c_H) \text{ and } \Pi_F^{C+S} = (1 - \gamma)(v_H - c_H) \\ \text{Thus, } \Pi_i^0 &= \Pi_i^S < \Pi_i^C < \Pi_i^{C+S} \text{ for both } i = T, F.\end{aligned}$$

Assume again $\gamma < \frac{v_L - c_L}{v_H - c_H}$. But now additionally assume $\gamma(v_H - c_H) + c_H > v_L$. Then consider following intervals:

- $\lambda_T < \gamma(v_L - c_L) + c_L$
 $\Pi_T^0 = \gamma(v_L - c_L)$ and $\Pi_F^0 = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
 Thus, $\Pi_i^0 = \Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$.
- $\lambda_T \in (\gamma(v_L - c_L) + c_L, \gamma(v_H - c_H) + c_L)$
 $\Pi_T^0 = \lambda_T - c_L > \gamma(v_L - c_L)$ and $\Pi_F^0 = v_L - \lambda_T < (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
 Thus, $\Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_F^0 < \Pi_F^C$ and $\Pi_T^C < \Pi_T^0 < \Pi_T^S$, but $\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^C$, so the differences in payoffs due to SC are only distributional.
- $\lambda_T \in (\gamma(v_H - c_H) + c_L, v_L)$
 $\Pi_T^0 = \lambda_T - c_L > \gamma(v_L - c_L)$ and $\Pi_F^0 = v_L - \lambda_T < (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
 Thus, $\Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_F^0 < \Pi_F^C$ (so that $\Pi_F^0 < \Pi_F^C < \Pi_F^S = \Pi_F^{C+S}$) and $\Pi_T^0 > \Pi_T^S$ (so that $\Pi_T^C < \Pi_T^S = \Pi_T^{C+S} < \Pi_T^0$). Yet, $\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$.
- $\lambda_T \in (v_L, \gamma(v_H - c_H) + c_H)$
 $\Pi_T^0 = \Pi_F^0 = 0$;
 $\Pi_T^S = \gamma(v_H - c_H)$ and $\Pi_F^S = (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
 Thus, $\Pi_i^0 < \Pi_i^C < \Pi_i^S = \Pi_i^{C+S}$ for both $i = T, F$.

- $\lambda_T \in (\gamma(v_H - c_H) + c_H, v_H)$
 $\Pi_T^0 = \Pi_F^0 = 0$;
 $\Pi_T^S = \lambda_T - c_H > \gamma(v_H - c_H)$ and $\Pi_F^S = v_H - \lambda_T < (1 - \gamma)(v_H - c_H)$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
Thus, $\Pi_i^0 < \Pi_i^C < \Pi_i^{C+S}$ for both $i = T, F$. Moreover, $\Pi_T^S > \Pi_T^{C+S}$ and $\Pi_F^0 < \Pi_F^S < \Pi_F^{C+S}$,
but $\Pi_F^S ? \Pi_F^C$. Also $\sum_{i=T,F} \Pi_i^0 < \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$.
- $\lambda_T > v_H$
 $\Pi_T^0 = \Pi_F^0 = 0$;
 $\Pi_T^S = \Pi_F^S = 0$;
 $\Pi_T^C = \gamma(v_L - c_L)$ and $\Pi_F^C = (1 - \gamma)(v_L - c_L)$;
 $\Pi_T^{C+S} = \gamma(v_H - c_H)$ and $\Pi_F^{C+S} = (1 - \gamma)(v_H - c_H)$
Thus, $\Pi_i^0 = \Pi_i^S < \Pi_i^C < \Pi_i^{C+S}$ for both $i = T, F$.

Notice that

- for $\lambda_T < v_L$

$$\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$$

- for $v_L < \lambda_T < v_H$

$$\sum_{i=T,F} \Pi_i^0 < \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^S = \sum_{i=T,F} \Pi_i^{C+S}$$

- for $\lambda_T > v_H$

$$\sum_{i=T,F} \Pi_i^0 = \sum_{i=T,F} \Pi_i^S < \sum_{i=T,F} \Pi_i^C < \sum_{i=T,F} \Pi_i^{C+S}$$